The Heart as a Pump: what to say

Built from the same content as the deck and the sheet, in the order the slides reveal it. Each paragraph is one click. The cues tell you what is on the screen; the words are a starting point for what you say, so change them into your own voice as you rehearse. Timing marks are rough, for a 75-minute lecture.

Opening. Last week we finished with the autonomic nervous system and the adrenal medulla. Today we put both of them to work on one organ. By the end of class you should be able to explain how a heart cell is told to contract, why the heart beats on its own, how that beat shows up on an ECG, what happens inside the ventricle during one cycle, and what decides how much blood leaves the heart each minute. We begin with smooth muscle, partly because the blood vessels we study next week are made of it, and partly because it shows very clearly that there is more than one way to switch a muscle on.

SHEET On the sheet, the first page is the drawing with numbered pins and a label list. Tell them: write the name on the numbered line as each pin drops, then fill the card for that number (what it is, what it does, why it matters) on the card pages at the back. Then the steps, then the effect box. The QR squares open any picture full size on their phone.

A. Smooth muscle: how it contracts and what controls it

about 12 min in

YOU SHOULD BE ABLE TO **Smooth muscle contraction mechanism.** Trace smooth muscle contraction through calcium entry and release to calmodulin and myosin light chain kinase and explain why the mechanism is slower and more economical than in skeletal muscle. **Smooth muscle types and regulation.** Compare single unit and multi unit smooth muscle and list the stimuli that alter smooth muscle tone including stretch and local chemical signals.

SAY FIRST Let me start with a question about muscle. Skeletal muscle needs a motor neuron and a troponin switch before it can contract. Smooth muscle needs neither, and that one difference explains most of how it behaves.

Label the drawing: Smooth muscle cell: the calcium switch

CONTEXT BOX One smooth muscle cell, cut open. The membrane is the outline, the small Ca^{2+} store (SR) sits inside, and the contractile proteins are at the lower right. There is no troponin and no sarcomere, so the switch is different: it is on myosin, and it runs through an enzyme. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the voltage-gated Ca^{2+} channel. The electrical door: an L-type channel that opens when the membrane depolarizes. Lets Ca^{2+} in from the extracellular fluid, the main Ca^{2+} source for smooth muscle. Why it matters: calcium channel blockers shut only this door, so they relax vessels and lower blood pressure without stopping tissues that run on the receptor route. What triggers it: Anything that moves the voltage: stretch opening stretch-sensitive cation channels, a pacemaker slow wave arriving through gap junctions, high extracellular K^+, or a transmitter that opens a cation channel or closes a K^+ channel.

PIN 2 This is the receptor-operated Ca^{2+} channel. The chemical door: Ca^{2+} entry and SR release started by a ligand on a G protein-coupled receptor, not by voltage. Lets Ca^{2+} rise with no action potential at all (pharmacomechanical coupling). Why it matters: which door a tissue uses is set by the receptors and channels it was built with: stretch and pacemakers use the voltage door, hormones the receptor door, nerves usually both, since norepinephrine also closes K^+ channels and depolarizes the cell. What triggers it: Norepinephrine on α_1, acetylcholine on M_3, angiotensin II, oxytocin, endothelin: the receptor activates phospholipase C, IP_3 opens the SR Ca^{2+} channels, and receptor-operated channels in the membrane open.

PIN 3 This is the sarcoplasmic reticulum (smooth muscle). A small internal Ca^{2+} store, much less developed than in skeletal muscle. Releases Ca^{2+} through IP_3 receptors and ryanodine receptors when signaled. Why it matters: smooth muscle leans on extracellular Ca^{2+} because this store is small.

PIN 4 This is the calmodulin. The Ca^{2+}-binding protein of smooth muscle; it takes the place troponin has in striated muscle. Binds four Ca^{2+}, then activates myosin light chain kinase. Why it matters: regulation is on the thick filament (myosin), not on actin.

PIN 5 This is the myosin light chain kinase (MLCK). The enzyme switched on by Ca^{2+}-calmodulin. Puts a phosphate on the myosin light chain, which turns on myosin ATPase so crossbridges can cycle. Why it matters: this extra enzyme step is one reason smooth muscle contracts slowly.

PIN 6 This is the phosphorylated myosin light chain. Myosin with a phosphate on its regulatory light chain. Only phosphorylated myosin can bind actin and cycle. Why it matters: dephosphorylated myosin can stay attached in the latch state: tension held with almost no ATP.

PIN 7 This is the dense bodies. Protein anchors in the cytoplasm and on the membrane, the smooth muscle version of Z discs. Actin filaments attach here, so shortening pulls the whole cell in like a drawstring. Why it matters: no sarcomeres, no striations, and the cell can shorten far more than skeletal muscle.

PIN 8 This is the myosin light chain phosphatase. The enzyme that removes the phosphate from the myosin light chain. Ends crossbridge cycling as Ca^{2+} falls and calmodulin releases MLCK. Why it matters: signals that change this enzyme's activity change tone without changing Ca^{2+} (Ca^{2+} sensitization).

How it happens: Smooth muscle cell: the calcium switch

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Ca^{2+} rises, by one of two doors. Electrical door: the cell depolarizes (stretch, a pacemaker cell, or a neurotransmitter opening a cation channel) and voltage-gated L-type Ca^{2+} channels open; Ca^{2+} enters from the ECF. Chemical door: a neurotransmitter or hormone binds a G protein-coupled receptor (norepinephrine on α_1, acetylcholine on M_3, angiotensin II, oxytocin). The receptor activates phospholipase C, IP_3 is made, IP_3 opens Ca^{2+} channels on the SR, and ligand-gated (receptor-operated) channels in the membrane let more Ca^{2+} in. No action potential is needed.

STEP 2 Ca^{2+} binds calmodulin. Four Ca^{2+} per calmodulin, in the cytoplasm. This is the smooth muscle version of Ca^{2+} meeting troponin; there is no troponin here.

STEP 3 Ca^{2+}-calmodulin activates myosin light chain kinase. MLCK is an enzyme that was sitting idle; Ca^{2+}-calmodulin binds it and turns it on. Because this is an enzyme step, it takes time, which is one reason smooth muscle is slow to start.

STEP 4 MLCK phosphorylates the myosin light chain. MLCK moves a phosphate from ATP onto the regulatory light chain of myosin. That phosphate switches on myosin ATPase; unphosphorylated myosin cannot bind actin.

STEP 5 Crossbridges cycle and the cell shortens. Phosphorylated myosin binds actin and pulls it toward the dense bodies, so the whole cell draws in like a drawstring. Cycling is 10 to 100 times slower than skeletal muscle.

STEP 6 Ca^{2+} falls and myosin phosphatase turns it off. Ca^{2+} is pumped out of the cell and back into the SR; calmodulin lets go of MLCK. Myosin light chain phosphatase removes the phosphate, so cycling stops. If a crossbridge is still attached when the phosphate comes off it can stay attached (latch state), holding tension for almost no ATP. Signals that inhibit the phosphatase (through Rho kinase) raise tension without raising Ca^{2+}.

EFFECT The effect, what changes because of this: Slow, graded contraction that can be held for hours as tone. Two doors let Ca^{2+} in: the voltage door (stretch, pacemaker slow waves, high K^+, a depolarizing transmitter) and the receptor door (a ligand on a G protein-coupled receptor, IP_3, no action potential needed); single-unit muscle leans on the first, multi-unit on the second, and most tissues have both. Tone changes whenever Ca^{2+} entry changes through either door or whenever the phosphatase is turned up or down. That is how norepinephrine constricts an arteriole (α_1, Ca^{2+} up), epinephrine relaxes a bronchiole (β_2, cAMP inhibits MLCK), nitric oxide relaxes a vessel (cGMP lowers Ca^{2+} and activates the phosphatase), and calcium channel blockers lower blood pressure. Give them a minute to write the effect box in their own words before you move on.

Fill in: Why smooth muscle is slower and cheaper to run

LINE 1 Regulation is on myosin (thick filament), through an enzyme step, not a troponin switch on actin.

LINE 2 Myosin ATPase is slow: 10 to 100 times slower crossbridge cycling than skeletal muscle.

LINE 3 The latch state crossbridges stay attached after dephosphorylation, holding tension with almost no ATP.

LINE 4 Ca^{2+} mostly comes from the ECF, so contraction can be graded by how much Ca^{2+} enters.

Mini mind map: What changes smooth muscle tone?

CENTER Center of the map: Smooth muscle tone. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 Stretch. Myogenic response: stretch opens channels, the cell depolarizes and contracts. Keeps blood flow steady when pressure rises.

BRANCH 2 Local chemicals. In arterioles: low O_2, high CO_2, H^+, adenosine and nitric oxide relax it; histamine relaxes it too (H_1 on the endothelium releases nitric oxide, H_2 on the muscle), which is the hypotension of anaphylaxis; endothelin tightens it. The same histamine contracts airway and gut smooth muscle (H_1 on the muscle): bronchospasm in the same patient. Tissues set their own blood flow this way; the receptor and the tissue decide the direction.

BRANCH 3 Autonomic neurons. Norepinephrine and acetylcholine released from varicosities; the receptor decides contract or relax. α_1 contracts, β_2 relaxes vascular smooth muscle.

BRANCH 4 Hormones. Epinephrine, angiotensin II, vasopressin, oxytocin and others. Oxytocin on the uterus is a positive feedback example.

BRANCH 5 Single-unit vs multi-unit. Single-unit: cells joined by gap junctions contract together, some are pacemakers (gut, uterus, small vessels). Multi-unit: each cell has its own nerve supply, fine control (iris, airways, large arteries). Which kind decides whether the tissue beats as one.

Clicker to close section A

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A drug blocks myosin light chain kinase in vascular smooth muscle. What happens to the vessel, and why?

A. It constricts, because calmodulin is now free to bind actin

B. It relaxes, because myosin cannot be phosphorylated and cannot cycle

C. Nothing changes, because troponin still controls the thin filament

D. It constricts, because the phosphatase is also blocked

Key: B. No MLCK means no phosphate on the myosin light chain, so myosin ATPase stays off and the vessel relaxes.

Not A: Calmodulin binds Ca^{2+}, not actin.

Not C: Smooth muscle has no troponin.

Not D: The phosphatase is a different enzyme, and blocking it would raise tone, not lower it.

Set B. A tissue's smooth muscle cells all contract together even though only a few of them receive a nerve ending. Which kind of smooth muscle is this, and what makes it possible?

A. Multi-unit; each cell has its own varicosity

B. Single-unit; gap junctions pass the signal cell to cell

C. Multi-unit; dense bodies link the cells mechanically

D. Single-unit; every cell has receptor-operated channels

Key: B. Single-unit (visceral) smooth muscle is electrically coupled by gap junctions, so a few innervated or pacemaker cells drive the whole sheet.

Not A: Multi-unit cells are individually innervated and do not contract as a group.

Not C: Dense bodies anchor actin inside a cell; they do not carry signals between cells.

Not D: Receptor-operated channels let Ca^{2+} in; they do not couple neighbors.

Set C. Smooth muscle in the bladder wall holds tension for an hour using very little ATP. Which feature explains that?

A. Fast myosin ATPase

B. The latch state: crossbridges stay attached after the light chain is dephosphorylated

C. Large sarcoplasmic reticulum stores

D. Troponin locks the thin filament open

Key: B. In the latch state dephosphorylated myosin stays bound to actin, holding force with almost no ATP use.

Not A: Smooth muscle myosin ATPase is slow, not fast.

Not C: The smooth muscle SR is small.

Not D: There is no troponin in smooth muscle.

BRIDGE Remember the enzyme switch and the latch state; both of them come back next week when we talk about blood vessels.

B. Cardiac muscle, and the three muscle types side by side

about 34 min in

YOU SHOULD BE ABLE TO **Cardiac muscle properties.** Explain how intercalated discs and gap junctions allow the myocardium to act as a functional syncytium and state why cardiac muscle cannot be tetanized. **Comparison of the three muscle types.** Build a comparison of skeletal, cardiac, and smooth muscle by calcium source, regulatory protein, control, and speed of contraction.

SAY FIRST Now the muscle we really care about today. Here are two cardiac cells joined end to end, and most of what makes the heart work as a pump comes down to the junction between them.

Label the drawing: Two cardiac cells and the intercalated disc

CONTEXT BOX Two cardiac muscle cells end to end. The stepped dark line between them is the intercalated disc, with its gap junction channels and desmosome anchors. Inside the left cell, a T-tubule drops in from the membrane and meets the sarcoplasmic reticulum. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the intercalated disc. The stepped junction where one cardiac cell ends and the next begins. Holds the cells together and carries the electrical signal across. Why it matters: because of it the myocardium works as one unit, a functional syncytium.

PIN 2 This is the gap junction. Channels (connexons) in the intercalated disc that join the cytoplasm of two cells. Let ions and the action potential pass straight to the next cell. Why it matters: this is why one SA node signal spreads to every atrial cell; the ventricles form a second syncytium.

PIN 3 This is the desmosome. A strong protein anchor in the intercalated disc. Holds cells together through the pull of every beat. Why it matters: without it the cells would tear apart under cardiac force.

PIN 4 This is the T-tubule (cardiac). An inward fold of the cell membrane, larger than in skeletal muscle. Carries the action potential deep into the cell to the L-type Ca^{2+} channels. Why it matters: this is where the electrical signal first lets Ca^{2+} in.

PIN 5 This is the L-type Ca^{2+} channel. A voltage-gated Ca^{2+} channel in the T-tubule membrane (the dihydropyridine receptor). Opens during the action potential plateau and lets a little Ca^{2+} in from the ECF. Why it matters: that trigger Ca^{2+} opens the ryanodine receptor: calcium-induced calcium release.

PIN 6 This is the ryanodine receptor (cardiac SR). The Ca^{2+} release channel on the sarcoplasmic reticulum. Opens when trigger Ca^{2+} binds it, releasing the SR store, about 90% of the Ca^{2+} for contraction. Why it matters: cardiac EC coupling needs both ECF and SR Ca^{2+}; skeletal muscle needs SR only.

PIN 7 This is the mitochondrion (cardiac). Up to a third of the cardiac cell's volume is mitochondria. Makes ATP aerobically, all day, from fatty acids, glucose and lactate. Why it matters: the heart cannot run an oxygen debt; cut its oxygen and the muscle dies (infarction).

PIN 8 This is the branching, single nucleus. Cardiac cells are short, branched and usually have one central nucleus. The branches let each cell touch several neighbors. Why it matters: this is how you tell cardiac muscle from skeletal (long cells with many nuclei at the edge) and from smooth (spindle-shaped cells with one nucleus and no striations).

How it happens: Two cardiac cells and the intercalated disc

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 An action potential arrives from the neighbor. Positive current flows through the gap junction channels (connexons) in the intercalated disc and depolarizes this cell to threshold. No nerve and no neurotransmitter are involved between cardiac cells.

STEP 2 The signal runs down the T-tubule. Deep into the cell, to the L-type Ca^{2+} channels.

STEP 3 Trigger Ca^{2+} enters. The L-type channel opens during the plateau and a little Ca^{2+} comes in from the ECF.

STEP 4 The SR releases the rest: calcium-induced calcium release. Trigger Ca^{2+} opens the ryanodine receptors and the store empties into the cytoplasm.

STEP 5 Both cells contract together, held by the desmosomes. Ca^{2+} binds troponin C, tropomyosin moves off the myosin-binding sites on actin, crossbridges cycle; force is graded by how much Ca^{2+} was released. The desmosomes keep the cells from tearing apart under that pull.

STEP 6 Ca^{2+} is pumped away and the cells relax. SERCA refills the SR and the Na^{+}-Ca^{2+} exchanger pushes Ca^{2+} out; mitochondria pay for all of it.

EFFECT The effect, what changes because of this: The whole chamber contracts as one unit, with force graded by how much Ca^{2+} was released. Lose the gap junctions and the heart cannot act as one; lose the oxygen supply and the mitochondria stop and the cells die, which is a heart attack. Give them a minute to write the effect box in their own words before you move on.

Fill the table: Compare the three muscle types

TABLE Across then down, one cell per click. Columns: Skeletal, Cardiac, Smooth.

ROW Ca^{2+} source. Skeletal: sarcoplasmic reticulum only. Cardiac: ECF through L-type channels, then the SR (calcium-induced calcium release). Smooth: mostly ECF, some SR.

ROW Regulatory protein. Skeletal: troponin, on the thin filament. Cardiac: troponin, on the thin filament. Smooth: calmodulin and MLCK, on the thick filament.

ROW Control. Skeletal: somatic motor neuron only; no nerve, no contraction. Cardiac: myogenic (SA node pacemaker); rate and force adjusted by autonomic nerves and hormones. Smooth: autonomic nerves, hormones, local chemicals, stretch; some pacemakers.

ROW Speed and tetanus. Skeletal: fastest; can summate and tetanize. Cardiac: intermediate; cannot tetanize (long refractory period). Smooth: slowest; sustained tone at low energy cost.

ROW Cell shape. Skeletal: long cylinder, many nuclei at the edge, striated. Cardiac: short, branched, one central nucleus, striated, intercalated discs. Smooth: spindle, one nucleus, no striations.

Clicker to close section B

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A toxin destroys the gap junctions in the intercalated discs but leaves the desmosomes intact. What happens to the heartbeat?

A. Nothing; the SA node still reaches every cell through the nerves

B. The cells stay attached but no longer fire together, so the myocardium stops acting as one unit

C. The cells pull apart with the first beat

D. Only the atria are affected

Key: B. Gap junctions carry the electrical signal cell to cell; without them the functional syncytium is lost even though the desmosomes keep the cells attached.

Not A: Cardiac cells are not individually innervated; the signal travels through gap junctions.

Not C: Desmosomes still hold the cells together.

Not D: Both atria and ventricles depend on gap junctions.

Set B. Which pairing of muscle type and calcium source is correct?

A. Skeletal: mostly extracellular Ca^{2+}

B. Cardiac: extracellular trigger Ca^{2+} then SR release

C. Smooth: sarcoplasmic reticulum only

D. Cardiac: sarcoplasmic reticulum only

Key: B. Cardiac muscle uses calcium-induced calcium release: a little Ca^{2+} from the ECF opens the SR store.

Not A: Skeletal muscle gets its Ca^{2+} from the SR alone.

Not C: Smooth muscle relies mostly on extracellular Ca^{2+}.

Not D: Cardiac muscle needs the extracellular trigger; a Ca^{2+}-free bath stops it.

Set C. A student says the heart could be tetanized if you stimulated it fast enough, like skeletal muscle. What is the best correction?

A. Correct; a fast enough stimulus tetanizes any muscle

B. Wrong; the cardiac refractory period lasts almost as long as the twitch, so a second contraction cannot stack on the first

C. Wrong; cardiac muscle has no sarcoplasmic reticulum to release more Ca^{2+}

D. Wrong; the fibrous skeleton insulates the cells from repeated stimuli

Key: B. The long plateau keeps Na^+ channels inactivated through most of the contraction, so summation and tetanus are impossible.

Not A: Skeletal muscle tetanizes because its refractory period is short; cardiac muscle's is not.

Not C: Cardiac muscle has an SR.

Not D: The fibrous skeleton separates atria from ventricles; it does not shield cells from stimuli.

BRIDGE So cardiac muscle acts as one unit and it cannot be tetanized. Keep both of those in mind when we get to the action potential.

C. The adrenal medulla: a sympathetic ganglion that secretes into blood

about 62 min in

YOU SHOULD BE ABLE TO **Adrenal medulla.** Explain why the adrenal medulla is a modified sympathetic ganglion and compare the duration of its circulating catecholamine effect with direct sympathetic innervation.

SAY FIRST This section is short, because it finishes something from last week: why the adrenal medulla counts as part of the sympathetic system, and why its signal lasts so much longer than a nerve signal.

Steps in order: From the spinal cord to every tissue

STEP 1 Sympathetic preganglionic neuron fires. Leaves the thoracic cord and releases acetylcholine onto nicotinic receptors, like any sympathetic preganglionic neuron.

STEP 2 Chromaffin cells respond. These are modified postganglionic neurons with no axons; that is why the medulla counts as a ganglion.

STEP 3 Catecholamines enter the blood. About 80% epinephrine, 20% norepinephrine, released as hormones.

STEP 4 Hormone reaches every tissue with adrenergic receptors. Including cells with no sympathetic nerve at all, such as skeletal muscle arterioles with β_2 receptors and liver cells.

STEP 5 Effect lasts minutes, not milliseconds. Cleared slowly by the liver; a sympathetic nerve signal ends in milliseconds by reuptake at the varicosity.

Fill in: Nerve versus hormone

LINE 1 Direct sympathetic nerve: speed and reach milliseconds, one target tissue, ended by reuptake.

LINE 2 Adrenal medulla: speed and reach seconds to start, whole body, lasts minutes.

LINE 3 Why both? the nerve gives fast local control; the hormone gives the whole-body fight or flight response and reaches tissues no nerve supplies.

Clicker to close section C

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A patient has a tumor of the adrenal medulla that dumps catecholamines into the blood in bursts. Why do the episodes last many minutes, when a sympathetic nerve response is over in milliseconds?

A. Hormones in blood are cleared slowly by the liver; nerve-released norepinephrine is taken back up at once

B. The tumor also fires the sympathetic nerves

C. Epinephrine binds receptors permanently

D. Blood flow to tissues is slower than nerve conduction

Key: A. Circulating catecholamines are removed slowly by liver enzymes, so the effect persists; at a synapse reuptake ends the signal in milliseconds.

Not B: The tumor secretes hormone; it does not drive the nerves.

Not C: Hormone binding to a receptor is reversible, so this cannot explain a long effect.

Not D: Delivery speed changes onset, not duration.

Set B. Why is the adrenal medulla called a modified sympathetic ganglion?

A. It receives postganglionic sympathetic axons

B. Its chromaffin cells are postganglionic neurons without axons, driven by preganglionic cholinergic neurons

C. It releases acetylcholine into the blood

D. It sits next to the sympathetic chain

Key: B. Preganglionic sympathetic neurons release ACh onto chromaffin cells, which are developmentally postganglionic neurons that secrete into blood instead of onto a target.

Not A: The medulla receives preganglionic axons, not postganglionic ones; that is the point of calling it a ganglion.

Not C: It releases epinephrine and norepinephrine.

Not D: Location is not the reason; its cell type and wiring are.

Set C. Skeletal muscle arterioles dilate during a fight or flight response even though they have almost no sympathetic nerve supply. Which signal reaches them?

A. Acetylcholine from the vagus

B. Epinephrine from the adrenal medulla acting on β_2 receptors

C. Norepinephrine from varicosities acting on α_1 receptors

D. Local nitric oxide only

Key: B. Circulating epinephrine reaches tissues no nerve reaches; on β_2 receptors it relaxes vascular smooth muscle.

Not A: The vagus does not supply skeletal muscle arterioles.

Not C: α_1 would constrict, and these vessels have little sympathetic innervation.

Not D: Nitric oxide is local and not part of the fight or flight signal.

BRIDGE So the sympathetic system has two ways to reach the body: a fast local wire and a slower whole-body broadcast through the blood. The heart answers to both.

D. The pacemaker and the cardiac action potential

about 104 min in

YOU SHOULD BE ABLE TO **Pacemaker potential.** Explain the unstable pacemaker potential of the sinoatrial node by the funny current, calcium entry, and potassium exit and state what sets intrinsic heart rate. **Contractile cell action potential.** Diagram the ventricular action potential and identify the ion movement responsible for the rapid upstroke, the plateau, and repolarization. **Cardiac refractory period.** Explain why the long refractory period of cardiac muscle prevents tetanus and why that matters for pump function.

SAY FIRST Why does the heart beat on its own, with no nerve telling it to? Because the SA node cell never rests. Let me show you its membrane potential over two beats.

Label the drawing: SA node cell: pacemaker potential

CONTEXT BOX Membrane potential of one SA node cell over two heartbeats. Notice there is no flat resting line: as soon as one beat ends the cell starts drifting toward the next. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the pacemaker potential (slow depolarization). The slow upward drift from about −60 mV to threshold between beats. Makes the SA node fire on its own, with no nerve input. Why it matters: its slope sets heart rate: steeper slope, faster heart.

PIN 2 This is the funny current, I_f. A current through HCN channels that open when the cell hyperpolarizes (the opposite of most channels). Lets Na^+ in, starting the pacemaker potential. Why it matters: sympathetic cAMP opens more of these channels; the drug ivabradine blocks them to slow the heart.

PIN 3 This is the T-type Ca^{2+} channels (SA node). Transient Ca^{2+} channels that open as the cell nears threshold. Add Ca^{2+} entry that finishes the climb to threshold. Why it matters: pacemaker depolarization is driven by Na^+ and Ca^{2+}, not by Na^+ alone.

PIN 4 This is the threshold in the SA node. About −40 mV in a pacemaker cell. Reaching this voltage opens the L-type Ca^{2+} channels. Why it matters: the SA node reaches threshold first (about 70 times a minute with vagal tone), so it sets the rhythm.

PIN 5 This is the upstroke of the SA node action potential. The rapid rise to about 0 mV. The rise is carried by Ca^{2+} entering through L-type channels, not by Na^+. Why it matters: pacemaker cells have no fast Na^+ channels, so the upstroke is slower than in a ventricular cell.

PIN 6 This is the repolarization of the SA node cell. The fall back toward −60 mV. K^+ leaves through voltage-gated K^+ channels as the Ca^{2+} channels close. Why it matters: as soon as the K^+ channels close, I_f starts the next pacemaker potential.

PIN 7 This is the minimum potential (about −60 mV). The lowest point of the cycle; there is no true resting potential. Hyperpolarization here is what opens the HCN channels. Why it matters: parasympathetic K^+ channels push this lower, so the

climb to threshold takes longer: slower heart.

How it happens: SA node cell: pacemaker potential

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 At −60 mV the funny channels open. HCN channels are unusual: they open when the membrane is hyperpolarized, not depolarized, and cAMP makes them open more. They pass mostly Na^+ in (a little K^+ out), so the cell drifts upward: the funny current, I_f.

STEP 2 T-type Ca^{2+} channels add to the climb. As the cell nears threshold the small, transient (T-type) voltage-gated Ca^{2+} channels open briefly and Ca^{2+} entry finishes the pacemaker potential.

STEP 3 Threshold, about −40 mV. The long-lasting (L-type) voltage-gated Ca^{2+} channels open; these are the same channel type a calcium channel blocker acts on.

STEP 4 Upstroke: Ca^{2+} in. Depolarization to about 0 mV carried by Ca^{2+}, not Na^+; there are no fast Na^+ channels here.

STEP 5 Voltage-gated K^+ channels open: repolarization. The L-type Ca^{2+} channels inactivate, delayed-rectifier K^+ channels open, K^+ leaves down its gradient and the potential falls back to −60 mV.

STEP 6 Back at −60 mV, it starts again. K^+ channels close, the funny channels reopen; this is the next beat starting.

EFFECT The effect, what changes because of this: The slope of the pacemaker potential sets heart rate. Sympathetic cAMP steepens it (faster); parasympathetic ACh flattens it and drops the floor (slower). That is the whole mechanism of heart rate control, and the SA node wins because its slope is steepest. Give them a minute to write the effect box in their own words before you move on.

Label the drawing: Ventricular contractile cell: action potential and twitch

CONTEXT BOX One ventricular contractile cell: membrane potential in solid color, the twitch tension dashed, and the refractory period as the bar underneath. Numbers on the left are millivolts. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the phase 4: resting membrane potential (ventricle). A stable −90 mV between beats. Held by K^+ leak; contractile cells wait for a signal from a neighbor. Why it matters: unlike the SA node, contractile cells do not drift to threshold on their own.

PIN 2 This is the phase 0: rapid upstroke. Depolarization to about +20 mV in 1 to 2 ms. Voltage-gated Na^+ channels open; Na^+ rushes in. Why it matters: fast upstroke means fast conduction from cell to cell.

PIN 3 This is the phase 1: brief early repolarization. A small dip just after the peak. Na^+ channels inactivate and a few K^+ channels open briefly. Why it matters: it sets up the plateau that comes next.

PIN 4 This is the phase 2: plateau. The long flat part, about 200 ms, near 0 mV. Ca^{2+} in through L-type channels balances K^+ out; this Ca^{2+} triggers contraction. Why it matters: the plateau makes the cardiac action potential as long as the twitch, which prevents tetanus.

PIN 5 This is the phase 3: repolarization (ventricle). The return to −90 mV. Ca^{2+} channels close, slow K^+ channels open, K^+ leaves. Why it matters: on the ECG this is the T wave.

PIN 6 This is the absolute refractory period. About 250 ms during which no second action potential can start, because the Na^+ channels are inactivated. It lasts almost as long as the contraction itself. Why it matters: no summation, no tetanus: the ventricle relaxes and fills between beats, which is the whole point of a pump.

PIN 7 This is the tension curve (cardiac twitch). The contraction that follows the action potential. Rises and falls while the cell is still refractory. Why it matters: in skeletal muscle the refractory period is over before tension peaks, so skeletal muscle can be tetanized and cardiac cannot.

How it happens: Ventricular contractile cell: action potential and twitch

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Phase 4: resting at −90 mV. K^+ leak holds it; this cell waits for a neighbor, it does not drift.

STEP 2 Phase 0: Na^+ rushes in. A neighbor depolarizes this cell to threshold through gap junctions; fast voltage-gated Na^+ channels open and Na^+ pours in; the upstroke takes 1 to 2 ms.

STEP 3 Phase 1: a brief dip. Na^+ channels inactivate and a few K^+ channels open.

STEP 4 Phase 2: the plateau, Ca^{2+} in. Voltage-gated L-type Ca^{2+} channels (opened by the depolarization) let Ca^{2+} in while K^+ permeability is low; the two currents balance for about 200 ms. This Ca^{2+} is the trigger for calcium-induced calcium release and the contraction.

STEP 5 Phase 3: K^+ out, repolarization. The Ca^{2+} channels inactivate and the slow delayed-rectifier K^+ channels open; K^+ leaves and the cell returns to −90 mV. On the ECG this is the T wave.

STEP 6 Refractory through most of the twitch. Na^+ channels stay inactivated about 250 ms, so no second action potential can start until the contraction is nearly over.

EFFECT The effect, what changes because of this: The plateau does two jobs: it brings in the Ca^{2+} that starts the contraction, and it keeps the cell refractory until the twitch is almost finished. So cardiac muscle cannot be tetanized, the ventricle always relaxes and refills, and the pump keeps pumping. Give them a minute to write the effect box in their own words before you move on.

Fill in: Rates and the refractory period

LINE 1 SA node intrinsic rate about 90 to 100 a minute on its own; about 70 with resting vagal tone.

LINE 2 AV node backup rate 40 to 60 a minute.

LINE 3 Purkinje fiber backup rate 25 to 40 a minute.

LINE 4 Why the heart cannot tetanize the refractory period (about 250 ms) lasts almost as long as the twitch, so a second contraction cannot stack on the first.

LINE 5 Why that matters the ventricle must relax to fill; a tetanized heart would stop pumping.

Clicker to close section D

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. A drug blocks HCN (funny current) channels in the SA node. What changes on the pacemaker potential, and what happens to heart rate?

A. The upstroke slows; rate rises

B. The slow depolarization is shallower, threshold takes longer to reach; rate falls

C. The minimum potential rises; rate rises

D. Repolarization is faster; rate is unchanged

Key: B. I_f drives the early pacemaker potential; block it and the drift to threshold is slower, so fewer beats per minute (this is how ivabradine works).

Not A: The upstroke is L-type Ca^{2+}, not I_f.

Not C: Blocking I_f does not change the minimum potential.

Not D: Repolarization is K^+ out, unrelated to I_f.

Set B. A ventricular cell is bathed in a solution with no Ca^{2+}. Which phase of its action potential disappears first, and what happens to the twitch?

A. Phase 0; the twitch gets stronger

B. Phase 2, the plateau; the twitch fails

C. Phase 3; the twitch lengthens

D. Phase 4; the cell fires faster

Key: B. The plateau is Ca^{2+} entry balancing K^+ exit; without Ca^{2+} there is no plateau and no trigger for SR release, so no contraction.

Not A: Phase 0 is Na^+ entry.

Not C: Phase 3 is K^+ exit.

Not D: Contractile cells do not fire on their own.

Set C. Which statement correctly contrasts the pacemaker cell with the contractile cell?

A. Pacemaker upstroke is Na^+; contractile upstroke is Ca^{2+}

B. Pacemaker cells have a stable −90 mV rest; contractile cells drift

C. Pacemaker upstroke is Ca^{2+} and there is no stable resting potential; contractile upstroke is Na^+ from a stable −90 mV

D. Both have a 200 ms plateau

Key: C. SA node cells lack fast Na^+ channels, use L-type Ca^{2+} for the upstroke, and never rest; contractile cells rest at −90 mV and fire with Na^+.

Not A: This option has the two cell types the wrong way around.

Not B: This option has the two cell types the wrong way around.

Not D: Only the contractile cell has the plateau.

BRIDGE The slope of the pacemaker potential sets the heart rate. Everything the nervous system does to heart rate, it does by changing that slope.

E. The wiring and the ECG

about 158 min in

YOU SHOULD BE ABLE TO **Conduction system.** Trace an impulse through the sinoatrial node, internodal pathways, atrioventricular node, bundle of His, bundle branches, and Purkinje fibers and explain the purpose of the atrioventricular delay. **ECG waves and intervals.** Label the P wave, QRS complex, T wave, PR interval, and QT interval and state the electrical event each represents. **ECG interpretation.** Calculate heart rate from an ECG strip and identify sinus rhythm, tachycardia, bradycardia, and a first degree or complete heart block. **ECG recording.** Record a lead II ECG, measure the intervals, and correlate each wave with the mechanical event that follows it.

SAY FIRST One signal from the SA node has to reach every ventricular cell, in the right order and at the right time. First we trace the wiring that carries it, and then we look at the same signal the way a nurse sees it, from the skin, on an ECG.

Label the drawing: Conduction system of the heart

CONTEXT BOX The heart from the front, so the patient's right is on your left. Pacemaker and conducting tissue are in color; the dashed line is the septum and the faint curve divides atria from ventricles. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the sinoatrial (SA) node. A patch of pacemaker cells in the right atrial wall near the superior vena cava. Fires first, about 70 times a minute, and starts every normal beat. Why it matters: a rhythm set by the SA node is called sinus rhythm.

PIN 2 This is the internodal pathways. Conducting fibers running through the atria from SA node to AV node. Spread the signal across both atria (P wave) and deliver it to the AV node. Why it matters: the fibrous skeleton stops the signal there: the

AV node is the only electrical road to the ventricles.

PIN 3 This is the atrioventricular (AV) node. Pacemaker and conducting tissue at the floor of the right atrium. Slows the signal by about 0.1 s so the atria finish contracting before the ventricles start. Why it matters: the AV delay is the PR segment; its backup rate is 40 to 60 a minute.

PIN 4 This is the bundle of His (AV bundle). The conducting bundle that pierces the fibrous skeleton into the septum. Carries the signal from the AV node to the bundle branches. Why it matters: it is the only electrical route through the fibrous skeleton, which otherwise insulates the atria from the ventricles.

PIN 5 This is the bundle branches. Left and right branches running down the interventricular septum. Carry the signal fast to the apex. Why it matters: depolarization starts at the apex, so the ventricles squeeze from the bottom up toward the outflow valves.

PIN 6 This is the purkinje fibers. Large, fast-conducting fibers that spread from the apex through the ventricular walls. Deliver the signal to the contractile cells almost at once (QRS lasts under 0.12 s). Why it matters: if every pacemaker above them fails, the Purkinje fibers can still drive the ventricles at 25 to 40 beats a minute.

How it happens: Conduction system of the heart

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 The SA node fires. The pacemaker with the steepest slope sets the pace, about 70 a minute.

STEP 2 The atria depolarize. Along the internodal pathways and cell to cell through gap junctions: the P wave; atrial contraction follows.

STEP 3 The AV node holds the signal about 0.1 s. The only road through the fibrous skeleton is slow on purpose, so the atria finish emptying first.

STEP 4 Bundle of His, then the bundle branches. Fast conduction down the septum toward the apex.

STEP 5 Purkinje fibers reach every ventricular cell. Within about 0.1 s the whole ventricle depolarizes: the QRS.

STEP 6 The ventricles squeeze from the apex up. Blood is pushed toward the valves at the top; then the ventricles repolarize: the T wave.

EFFECT The effect, what changes because of this: Atria first, a pause, then the ventricles from the bottom up: that order is what makes the pump efficient. Every wave on the ECG is one of these steps seen from the skin, and a block at any point shows up as a stretched or missing connection between P and QRS. Give them a minute to write the effect box in their own words before you move on.

Label the drawing: Lead II, one beat

CONTEXT BOX Lead II, one beat, on standard ECG paper. A small box is 0.04 s, a big box 0.20 s. The flat line is the baseline; every wave is read against it. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the P wave. The first small bump. It records the atria depolarizing as the signal spreads out from the SA node. Why it matters: atrial contraction follows it; a P wave before every QRS is one mark of sinus rhythm.

PIN 2 This is the QRS complex. The tall spike. It records the ventricles depolarizing; the atria repolarize at the same time, but that small signal is hidden inside the QRS. Why it matters: ventricular contraction follows; wider than 0.12 s means a slow route through the ventricles.

PIN 3 This is the T wave. The rounded wave after the QRS. It records the ventricles repolarizing. Why it matters: ventricular relaxation follows it.

PIN 4 This is the PR interval. Start of P to start of QRS, normally 0.12 to 0.20 s. Measures conduction from the atria through the AV node. Why it matters: longer than 0.20 s, every beat, is first-degree heart block.

PIN 5 This is the QT interval. Start of QRS to end of T. Covers ventricular depolarization and repolarization: one ventricular action potential. Why it matters: a long QT raises the risk of dangerous ventricular arrhythmia.

PIN 6 This is the ST segment. The flat stretch between the S wave and the T wave. During this flat stretch every ventricular cell is sitting at its plateau, fully depolarized. Why it matters: lifted or dropped above or below baseline points to injured heart muscle.

PIN 7 This is the isoelectric baseline. The flat line between beats. No net electrical movement: the ventricles are at rest. Why it matters: every wave is measured up or down from this line.

How it happens: Lead II, one beat

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 P wave: the atria depolarize. The SA node signal spreading across both atria; atrial contraction follows.

STEP 2 PR segment: the AV node delay. Flat line while the signal waits in the AV node; PR interval normally 0.12 to 0.20 s.

STEP 3 QRS: the ventricles depolarize. Fast, under 0.12 s, because of the Purkinje system; ventricular contraction follows.

STEP 4 ST segment: all depolarized. The plateau of every ventricular cell at once; a shifted ST means injured muscle.

STEP 5 T wave: the ventricles repolarize. Phase 3 of every cell; ventricular relaxation follows.

STEP 6 Baseline: rest and filling. Nothing moving electrically; the ventricles are filling for the next beat.

EFFECT The effect, what changes because of this: The ECG is the electrical story of one beat, and each mechanical event follows its wave by a fraction of a second. Rate comes from the spacing of the R waves; a stretched PR is first-degree block; P waves marching on their own with slow wide QRS complexes is complete block. Give them a minute to write the effect box in their own words before you move on.

Fill in: Read the strip

LINE 1 Heart rate from the strip 1500 ÷ small boxes between two R waves, or 300 ÷ big boxes; or count beats in 6 seconds × 10.

LINE 2 Sinus rhythm a P wave before every QRS, a QRS after every P, regular, 60 to 100 a minute.

LINE 3 Tachycardia sinus rhythm faster than 100 a minute.

LINE 4 Bradycardia sinus rhythm slower than 60 a minute.

LINE 5 First-degree heart block every P is followed by a QRS, but the PR interval is longer than 0.20 s.

LINE 6 Complete (third-degree) heart block P waves and QRS complexes march independently; the ventricles run on a Purkinje escape rate of 25 to 40.

LINE 7 Wave to mechanical event P then atrial contraction; QRS then ventricular contraction; T then ventricular relaxation.

Clicker to close section E

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. On a strip there are 25 small boxes between R waves, and every P is followed by a QRS with a PR of 0.16 s. Rate and rhythm?

A. 60 a minute, sinus rhythm

B. 75 a minute, first-degree block

C. 100 a minute, sinus tachycardia

D. 60 a minute, sinus bradycardia

Key: A. 1500 ÷ 25 = 60; P before every QRS and a normal PR is sinus rhythm, and 60 is the lower edge of normal, not bradycardia.

Not B: PR of 0.16 s is normal; block needs more than 0.20 s.

Not C: 100 would be 15 small boxes.

Not D: Bradycardia means a rate below 60, and this rate is exactly 60.

Set B. A strip shows P waves at 80 a minute and wide QRS complexes at 35 a minute with no fixed relationship between them. What is going on?

A. Sinus bradycardia

B. First-degree heart block

C. Complete heart block; the ventricles are on a Purkinje escape rhythm

D. Sinus tachycardia with a long QT

Key: C. Independent P and QRS means nothing gets through the AV node; a slow, wide ventricular rhythm at 25 to 40 is a Purkinje escape.

Not A: The atria are at 80, not slow.

Not B: First-degree block still conducts every P with a long PR.

Not D: Neither the atria nor the ventricles are beating fast in this strip.

Set C. Suppose a drug removed the AV node delay. What would go wrong with the pump?

A. Nothing; the ventricles would just beat sooner

B. The ventricles would contract while the atria are still emptying into them, so ventricular filling drops

C. The atria would stop contracting

D. The QRS would widen

Key: B. The 0.1 s delay lets atrial contraction finish topping up the ventricles before they squeeze.

Not A: Timing is the point of the delay.

Not C: Atrial contraction is driven by the SA node, not the AV delay.

Not D: QRS width depends on the ventricular conduction route, not the AV delay.

BRIDGE So we have P, QRS and T, and each contraction follows its wave by a fraction of a second. Now let us look at what the ventricle is actually doing mechanically during all of this.

F. The cardiac cycle, the loop, and the sounds

about 224 min in

YOU SHOULD BE ABLE TO **Cardiac cycle.** Order the phases of the cardiac cycle and state the pressure relationship that opens and closes the atrioventricular and semilunar valves in each phase. **Pressure volume loop.** Label the four phases of a ventricular pressure volume loop and predict how the loop changes with increased preload, increased afterload, or increased contractility. **Heart sounds.** Explain the origin of the first and second heart sounds and relate a murmur to valve stenosis or regurgitation by its timing. **Heart sound and pulse correlation.** Auscultate the heart sounds, locate the valve areas, and correlate the sounds with the pulse and with the cardiac cycle phases.

SAY FIRST The cardiac cycle has five phases, four valve events, and two sounds you can hear with a stethoscope. The pressure volume loop is the same cycle drawn as pressure against volume, which makes the valve events easy to see.

Steps in order: The cardiac cycle, phase by phase

STEP 1 Atrial systole. Atria contract and top up the ventricles to EDV; AV valves open, semilunar valves closed.

STEP 2 Isovolumic ventricular contraction. Ventricular pressure rises above atrial pressure: AV valves close, S1. Not yet above aortic pressure, so all valves closed and volume fixed.

STEP 3 Ventricular ejection. Ventricular pressure passes aortic and pulmonary pressure: semilunar valves open, blood leaves, volume falls to ESV.

STEP 4 Isovolumic ventricular relaxation. Pressure falls below aortic pressure: semilunar valves close, S2. Still above atrial pressure, so all valves closed.

STEP 5 Ventricular filling. Pressure falls below atrial pressure: AV valves open, blood pours in passively; most filling happens here.

Label the drawing: Left ventricular pressure volume loop

CONTEXT BOX Left ventricular pressure (up) against volume (across) for one beat. Time runs counterclockwise around the loop; the corners are the four valve events. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the ventricular filling (PV loop). The bottom of the loop, running left to right. Volume rises from ESV to EDV while pressure stays low; mitral valve open. Why it matters: most filling is passive; atrial contraction adds the last part.

PIN 2 This is the point A: end-diastolic volume, mitral valve closes. The right-hand bottom corner, about 135 mL. Ventricular pressure rises above atrial pressure, the mitral valve closes: S1. Why it matters: eDV is the preload; raise it and the loop widens.

PIN 3 This is the isovolumic contraction. The straight line up on the right. Pressure rises with all valves closed, so volume cannot change. Why it matters: pressure must beat aortic pressure (about 80 mm Hg) before anything leaves.

PIN 4 This is the point B: aortic valve opens. The top of the right side, about 80 mm Hg. Ventricular pressure passes aortic pressure and ejection begins. Why it matters: the afterload is the pressure the ventricle has to reach here; raise it and the loop gets taller and narrower.

PIN 5 This is the ventricular ejection. The top of the loop, running right to left. Volume falls from EDV to ESV as blood leaves; pressure peaks near 120 mm Hg. Why it matters: the horizontal width of this part is the stroke volume.

PIN 6 This is the point C: end-systolic volume, aortic valve closes. The top left corner, about 65 mL. Ventricular pressure drops below aortic pressure; the aortic valve closes: S2. Why it matters: eSV is what is left; stronger contractility pushes this corner left.

PIN 7 This is the isovolumic relaxation. The straight line down on the left. Pressure falls with all valves closed. Why it matters: it ends the moment ventricular pressure drops below atrial pressure, which opens the mitral valve.

PIN 8 This is the point D: mitral valve opens. The bottom left corner. Ventricular pressure falls below atrial pressure, the mitral valve opens, filling starts again. Why it matters: the loop runs counterclockwise, one lap per beat.

How it happens: Left ventricular pressure volume loop

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Filling: mitral open, pressure low. Volume climbs from ESV to EDV, most of it passive, the last bit from atrial contraction.

STEP 2 A: the mitral valve closes. Ventricular pressure rises above atrial pressure: S1. Volume here is EDV, the preload.

STEP 3 Isovolumic contraction. All valves closed, pressure climbs, volume cannot change.

STEP 4 B: the aortic valve opens. Ventricular pressure passes aortic pressure, about 80 mm Hg: the afterload.

STEP 5 Ejection. Blood leaves, volume falls from EDV to ESV, pressure peaks near 120 mm Hg.

STEP 6 C: the aortic valve closes. Pressure drops below aortic pressure: S2. Volume here is ESV.

STEP 7 Isovolumic relaxation. All valves closed again while pressure falls.

STEP 8 D: the mitral valve opens. Pressure drops below atrial pressure and filling starts over.

EFFECT The effect, what changes because of this: The width of the loop is the stroke volume. More preload moves A to the right and widens it (Frank-Starling). More afterload lifts B and narrows it. More contractility pulls C to the left and widens it. A failing ventricle makes a short, narrow loop. Give them a minute to write the effect box in their own words before you move on.

Mini mind map: What you hear, and when

CENTER Center of the map: Heart sounds. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 S1, 'lub'. AV valves (mitral and tricuspid) closing at the start of ventricular systole. Loudest at the apex.

BRANCH 2 S2, 'dup'. Semilunar valves (aortic and pulmonary) closing at the start of diastole. Loudest at the base, 2nd intercostal spaces.

BRANCH 3 Systolic murmur. Between S1 and S2: an AV valve leaking (regurgitation) or a semilunar valve too narrow (stenosis). Timing tells you which valves are candidates.

BRANCH 4 Diastolic murmur. After S2: a semilunar valve leaking or an AV valve too narrow. Mitral stenosis is the classic one.

BRANCH 5 The pulse. Felt just after S1, during ejection; one pulse per S1–S2 pair. Counting the pulse counts ventricular beats.

Label the drawing: Valve areas on the chest wall

CONTEXT BOX The front of the chest: sternum in the middle, ribs counted from the top, the heart faint behind. Sound travels in the direction the blood flows, so each valve is heard a little downstream of where it sits. First we name the parts. Then we follow what happens, step by step.

PIN 1 This is the aortic area. Second intercostal space, right sternal border. This is the best place to hear the aortic valve, and S2 is loud here. Why it matters: aortic stenosis gives a harsh systolic murmur here.

PIN 2 This is the pulmonic area. Second intercostal space, left sternal border. This is the best place to hear the pulmonary valve. Why it matters: sound travels in the direction of flow, which is why the valve areas sit above the valves themselves.

PIN 3 This is the tricuspid area. Fourth or fifth intercostal space, left lower sternal border. This is the best place to hear the tricuspid valve. Why it matters: tricuspid regurgitation gives a systolic murmur here.

PIN 4 This is the mitral area (apex). Fifth intercostal space, left midclavicular line: the apex. S1 is loudest here; this is also where you feel the apical pulse. Why it matters: mitral stenosis gives a diastolic murmur here; mitral regurgitation a systolic one.

How it happens: Valve areas on the chest wall

CONTEXT BOX This is the same drawing, and now we follow the sequence. Each click lights up where the step happens and draws a gold line to it, and the close-up in the box shows that moment. Have them write the step name in the numbered box on the sheet and the detail under it.

STEP 1 Aortic area: right 2nd space. S2 is loud here; aortic stenosis gives a harsh systolic murmur.

STEP 2 Pulmonic area: left 2nd space. The other S2 valve.

STEP 3 Tricuspid area: left lower sternal border. S1 from the right side; tricuspid regurgitation is a systolic murmur here.

STEP 4 Mitral area: 5th space, midclavicular. The apex; S1 loudest, and the apical pulse is felt here.

EFFECT The effect, what changes because of this: Listen at each site for the two sounds and anything between them. A murmur between S1 and S2 is systolic (an AV valve leaking or a semilunar valve too narrow); after S2 it is diastolic (the reverse). Feel the radial pulse while you listen: it arrives just after S1. Give them a minute to write the effect box in their own words before you move on.

Clicker to close section F

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. During which phase are all four valves closed and ventricular pressure rising?

A. Ventricular filling

B. Isovolumic contraction

C. Ventricular ejection

D. Atrial systole

Key: B. AV valves have just closed (S1) and pressure has not yet reached aortic pressure, so nothing moves and pressure climbs.

Not A: The AV valves are open during filling.

Not C: The semilunar valves are open during ejection.

Not D: The AV valves are open while the atria contract.

Set B. A murmur is heard between S1 and S2, loudest at the apex. Which valve problem fits?

A. Mitral stenosis

B. Aortic regurgitation

C. Mitral regurgitation

D. Pulmonic stenosis

Key: C. Between S1 and S2 is systole; a systolic murmur at the apex is blood leaking back through a mitral valve that should be shut.

Not A: Mitral stenosis murmurs are diastolic.

Not B: Aortic regurgitation is diastolic and heard at the base.

Not D: Pulmonic stenosis is systolic but at the left 2nd space, not the apex.

Set C. On a pressure volume loop, afterload rises (the patient's arterial pressure is high). How does the loop change?

A. Wider, because the ventricle fills more

B. Taller and narrower: point B is higher and less blood is ejected

C. Shifted right with no change in shape

D. Shorter, because the ventricle contracts less forcefully

Key: B. The ventricle must reach a higher pressure before the aortic valve opens (B moves up), and ejection ends sooner, so SV falls and ESV rises.

Not A: Afterload does not increase filling.

Not C: Shape changes; it does not simply slide.

Not D: Contractility is unchanged; the ventricle works against more pressure.

BRIDGE The width of the loop is the stroke volume. Everything in the last section is about what makes that width bigger or smaller.

G. Stroke volume, cardiac output, and what controls them

about 299 min in

YOU SHOULD BE ABLE TO **Stroke volume and ejection fraction.** Calculate stroke volume and ejection fraction from end diastolic and end systolic volumes and interpret a reduced ejection fraction. **Cardiac output.** Calculate cardiac output and cardiac reserve and predict the effect of a change in heart rate or stroke volume on each. **Frank Starling relationship.** State the Frank Starling law and explain how venous return and end diastolic volume determine the force of the next contraction. **Preload afterload and contractility.** Distinguish preload, afterload, and contractility and predict how a change in each alters stroke volume. **Autonomic regulation of the heart.** Explain how sympathetic and parasympathetic input alter heart rate and contractility at the receptor and second messenger level.

SAY FIRST Last section. We work out the numbers for one beat and one minute, and then we look at the three things that control them.

Fill in: The numbers (typical resting adult)

LINE 1 End-diastolic volume (EDV) about 135 mL.

LINE 2 End-systolic volume (ESV) about 65 mL.

LINE 3 Stroke volume = EDV – ESV 135 – 65 = 70 mL per beat.

LINE 4 Ejection fraction = SV ÷ EDV 70 ÷ 135 = 52%; under about 40% means the ventricle is failing to empty (reduced ejection fraction).

LINE 5 Cardiac output = HR × SV 72 × 70 mL = 5.0 L/min, about the whole blood volume every minute.

LINE 6 Cardiac reserve maximum CO minus resting CO; a trained heart can reach 20 to 25 L/min.

LINE 7 Change HR or SV CO rises or falls in proportion; raising both at once is how exercise gets a fourfold increase.

Mini mind map: What sets cardiac output?

CENTER Center of the map: Cardiac output = HR × SV. One branch per click; each branch has its own small drawing of how that one works. Have them draw the map on the sheet, with the branch name first and the explanation under the picture.

BRANCH 1 Heart rate: sympathetic up. Norepinephrine and epinephrine on β_1 receptors in the SA node speed the pacemaker potential. cAMP opens more I_f channels and Ca^{2+} channels.

BRANCH 2 Heart rate: parasympathetic down. Acetylcholine on M_2 receptors opens K^+ channels and slows the drift to threshold. Resting vagal tone is why we sit at 70, not 100.

BRANCH 3 Preload. End-diastolic volume, set by venous return; more stretch means more force (Frank-Starling). Raise venous return and SV rises on the next beat.

BRANCH 4 Contractility. Force at a given EDV; raised by sympathetic input and epinephrine through β_1 and more Ca^{2+}. Digoxin raises it too; β-blockers lower it.

BRANCH 5 Afterload. The arterial pressure the ventricle must beat to open the semilunar valve; higher afterload means lower SV at the same force. Hypertension and aortic stenosis raise it.

Steps in order: Sympathetic input at the receptor level

STEP 1 Norepinephrine or epinephrine binds β_1. A G protein-coupled receptor on pacemaker and contractile cells.

STEP 2 G_s activates adenylyl cyclase. Cytoplasmic cAMP rises.

STEP 3 In the SA node, cAMP acts on HCN channels. More I_f, steeper pacemaker potential, faster rate (chronotropy).

STEP 4 In contractile cells, protein kinase A phosphorylates targets. L-type Ca^{2+} channels let in more Ca^{2+}, more SR release, stronger contraction (inotropy).

STEP 5 PKA also phosphorylates phospholamban. SERCA pumps Ca^{2+} back faster, so relaxation is quicker and there is time to fill at high rates.

STEP 6 Parasympathetic does the reverse at the SA node. ACh on M_2 receptors: G_i opens K^+ channels (hyperpolarizes) and lowers cAMP; little direct effect on ventricular force.

Fill in: Frank–Starling in one line each

LINE 1 The law the more the ventricle is stretched by filling (EDV), the harder it contracts, so stroke volume matches venous return.

LINE 2 Why at longer sarcomere length more crossbridges can form and the filaments are more sensitive to Ca^{2+}.

LINE 3 What raises venous return skeletal muscle pump, respiratory pump, sympathetic venoconstriction, more blood volume.

LINE 4 Why it matters the two sides of the heart stay matched beat to beat without any nerve telling them to.

Clicker to close section G

CLICKER Launch the clicker. The question goes to their devices for 30 seconds; the screen shows only the room code and the clock. When it closes, the question, the bars and the key appear. Walk through why each option is right or wrong before you move on. The three class sets ask different questions; here are all three.

Set A. EDV 150 mL, ESV 90 mL, HR 80. Stroke volume, ejection fraction and cardiac output?

A. 60 mL, 40%, 4.8 L/min

B. 60 mL, 67%, 4.8 L/min

C. 90 mL, 60%, 7.2 L/min

D. 150 mL, 40%, 12 L/min

Key: A. SV = 150 − 90 = 60; EF = 60 ÷ 150 = 40%; CO = 80 × 60 = 4800 mL/min. An EF of 40% is low: this ventricle is not emptying well.

Not B: 67% would be 60 ÷ 90, the wrong denominator.

Not C: ESV is what stays behind, not what leaves.

Not D: EDV is not the stroke volume.

Set B. A patient's venous return rises suddenly when they lie down. Without any nerve input, what happens to the next stroke volume and why?

A. It falls, because afterload rises

B. It rises, because greater EDV stretches the fibers and they contract harder (Frank-Starling)

C. It is unchanged until the baroreflex acts

D. It rises, because heart rate increases

Key: B. Frank–Starling: more filling, more stretch, more force, more ejected; this matches output to return beat by beat with no nerve.

Not A: Lying down does not raise arterial pressure in that instant.

Not C: The length-tension response is intrinsic and immediate.

Not D: Rate is a separate variable and was not changed here.

Set C. A β_1 blocker is given. Predict the effect on heart rate and contractility, and name the step that is interrupted.

A. Both fall; norepinephrine can no longer raise cAMP in SA node and contractile cells

B. Rate falls, contractility rises; K^+ channels close

C. Both rise; the drug mimics epinephrine

D. Only contractility falls; the SA node has no β_1 receptors

Key: A. β_1 receptors on both cell types work through G_s and cAMP; block them and the pacemaker potential is less steep and less Ca^{2+} enters contractile cells.

Not B: Contractility falls too, and K^+ channels belong to the M_2 pathway.

Not C: A blocker prevents, it does not mimic.

Not D: The SA node is rich in β_1 receptors.

Closing. Today you saw three kinds of muscle, one pacemaker, the heart's wiring, one full cycle, and the one equation that ties it together: cardiac output equals heart rate times stroke volume. Next week the blood leaves the heart and we follow the pressure through the vessels and back again. Before then, work through the study page once with your sheet in front of you, and bring any questions about the parts that did not make sense yet.